United States Patent	12399439
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Chang; Hsuan Jung et al.

Pellicle, pellicle removal tool, and method of utilizing pellicle and pellicle removal tool

Abstract

A dis-pellicle tool or pellicle removal tool that includes one or more clamp structures that each include a respective group of pins. The respective pins of the groups of pins of the one or more clamp structures are configured to, in operation, be inserted into holes that are present within a pellicle frame of a pellicle. Once the respective pins of the groups of pins are inserted into the holes of the pellicle frame, a removal force may be applied to the pellicle frame to remove the pellicle from a photomask to which the pellicle is coupled to by an adhesive, glue, or some other similar type of coupling material.

Inventors: **Chang; Hsuan Jung (Hsinchu, TW), Hsieh; Kun-Lung (Hsinchu, TW), Ko; Ting-Hsien (Hsinchu, TW), Wen; Chih-Wei (Hsinchu, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.** (Hsinchu, TW)

Family ID: **1000008778849**

Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)**

Appl. No.: **18/319362**

Filed: **May 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240385544 A1	Nov. 21, 2024

Publication Classification

Int. Cl.: **G03F7/20** (20060101); **B32B43/00** (20060101); **G03F7/00** (20060101)

U.S. Cl.:

CPC **G03F7/70983** (20130101); **B32B43/006** (20130101);

Field of Classification Search

CPC: G03F (7/70983); B32B (43/006)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5953107	12/1998	Li	355/77	G03F 1/64
2005/0088637	12/2004	Laganza	430/5	G03F 7/70983
2011/0215602	12/2010	Shirasaki	294/119.1	B25B 9/02
2022/0026797	12/2021	Yoo et al.	N/A	N/A
2022/0334465	12/2021	Huang et al.	N/A	N/A
2022/0334494	12/2021	Zhao	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106687860	12/2016	CN	N/A
201339740	12/2012	TW	N/A
201837600	12/2017	TW	N/A

Primary Examiner: Asfaw; Mesfin T

Attorney, Agent or Firm: Seed IP Law Group LLP

Background/Summary

BACKGROUND

(1) In the manufacturing of integrated circuit devices, photomasks, alternatively referred to as photoreticles or reticles, are used to project patterns for the integrated circuits on a semiconductor wafer. To protect the photomask from contamination, which can cause errors in the projected patterns, the photomask has been provided with a pellicle comprised of a pellicle frame and an optically transparent membrane. An adhesive is used to bond the pellicle frame to a surface of the photomask, and the membrane extends across the pellicle frame.

(2) Reasons for removing the pellicle include repairing a defect detected in the photomask, removing haze contamination that has formed on the photomask under the pellicle, and replacing the pellicle due to mechanical damage or exposure damage. Haze contamination is a precipitant formed from chemical residue from the photomask cleaning, impurity of fab or environmental exposure. Currently, precipitated defects or contamination may be caused by airborne contamination from the environment, pellicle adhesive, photomask pod out-gassing, pellicle frame residue, chemical growth and deposition from the reactions, and mixing of chemical solutions.

(3) When the pellicle is removed, damage may occur to the membrane due to forces applied to a pellicle frame of the pellicle being applied in an unbalanced fashion resulting in flexure and bending in the membrane, which may damage the membrane. For example, this damage to the membrane may include cracking, wrinkling, or rupturing that propagates and occurs through and

within the membrane such that the membrane is no longer suitable to be utilized to protect a respective photomask or allow light of selected wavelengths to pass through it unimpeded.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1**A is a top plan view of a pellicle coupled to a photomask and one or more clamps of a dis-pellicle tool.

(3) FIG. **1**B is a cross-sectional view taken along line **1**B-**1**B′ as shown in FIG. **1**A of the pellicle coupled to the photomask and the one or more clamps of the dis-pellicle tool as shown in FIG. **1**A.

(4) FIG. **2** is a top plan view of a respective pellicle after removal from a respective photomask that resulted in damage and defects to propagate through and within a membrane of the pellicle.

(5) FIG. **3**A is a top plan view of a pellicle coupled to a photomask and one or more clamps of a dis-pellicle tool, in accordance with some embodiments.

(6) FIG. **3**B is a cross-sectional view taken along line **3**B-**3**B′ as shown in FIG. **3**A of the pellicle coupled to the photomask and the one or more clamps of the dis-pellicle tool as shown in FIG. **3**B, in accordance with some embodiments.

(7) FIG. **3**C is a side view of the pellicle coupled to the photomask and one or more clamps of the dis-pellicle tool as shown in FIGS. **3**A and **3**B, in accordance with some embodiments.

(8) FIG. **3**D is a zoomed in and enhanced view of section **3**D-**3**D′ of the pellicle as shown in FIG. **3**C, in accordance with some embodiments.

(9) FIG. **4**A is a top plan view of a profile of a pin along one or more clamps of a dis-pellicle tool, in accordance with some embodiments.

(10) FIG. **4**B is a front side view of an end of the pin along one or more clamps of the dis-pellicle tool as shown in FIG. **4**A, in accordance with some embodiments.

(11) FIG. **5**A is a top plan view of a system configured to, in operation, control and position one or more clamps of a dis-pellicle tool, in accordance with some embodiments.

(12) FIG. **5**B is a zoomed in and enhanced view of section **5**B-**5**B′ as shown in FIG. **5**A of at least one pin along a clamp of a dis-pellicle tool and at least one reception opening along a frame of a pellicle, in accordance with some embodiments.

(13) FIG. **6** is a cycle of removing a pellicle from a photomask and reutilizing the pellicle after removal from the photomask, in accordance with some embodiments.

(14) FIG. **7** is a flowchart of a method of removing a pellicle from a photomask to which the pellicle is coupled utilizing one or more clamps with one or more pins, in accordance with some embodiments.

(15) FIGS. **8**A-**8**E are respective views of respective steps of the embodiment of the method of removing the pellicle from the photomask to which the pellicle is coupled as shown in the flowchart of FIG. **7**.

DETAILED DESCRIPTION

(16) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which

additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(17) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(18) Generally, semiconductor wafers, substrates or layers are patterned during a manufacturing process utilizing an extreme ultra-violet (EUV) patterning tool or device. For example, a photomask, which is coupled to a pellicle such that the photomask is overlapped by the pellicle, is aligned with a semiconductor wafer within the EUV patterning tool. EUV light is then emitted by an EUV light source within the EUV patterning tool at the semiconductor wafer to pattern the semiconductor wafer.
(19) After this EUV patterning process is carried out one or more times, the pellicle may be removed from the photomask by a dis-pellicle or pellicle removal tool by utilizing one or more clamp structures of the dis-pellicle or pellicle removal tool. For example, the one or more clamp structures may be engaged with a pellicle frame of the pellicle and the one or more clamp structures may then be moved away from the photomask. As the one or more clamp structures apply the force to the pellicle, an adhesion force of an adhesive coupling the pellicle frame to the photomask is overcome resulting in the pellicle frame of the pellicle breaking away from the adhesive. However, when the force is applied in an unbalanced fashion such that the force is unevenly distributed across the pellicle frame, a membrane of the pellicle coupled to the pellicle frame may be damaged. For example, damage to the membrane may include wrinkling, rupturing, cracking, or some other similar type of damage that may occur in the membrane when the force is unevenly distributed across the membrane. When the membrane is damaged, which is configured to, in operation, protect the photomask from contamination when being utilized within the EUV patterning tool, the pellicle is disposed of resulting in increased manufacturing costs and waste costs as pellicles are relatively expensive.
(20) When this damage occurs in the membrane of the pellicle when removing the pellicle from the photomask, small particles of the membrane may break away from the membrane resulting in the photomask being damaged or contaminated. When the photomask is damaged or contaminated, the photomask is disposed of resulting in increased manufacturing and waste costs as photomasks are relatively expensive.
(21) The present disclosure is directed to providing one or more embodiments of a dis-pellicle tool or pellicle removal tool including one or more clamp structures that are utilized to apply a removal force to a pellicle frame of a pellicle in order to remove the pellicle from a photomask. When the force is applied by the one or more clamp structures to the pellicle frame, the force is applied in a balanced manner across the pellicle frame such that the force is evenly distributed across and along the pellicle frame. This even distribution of the removal force prevents or reduces the likelihood of damage to the pellicle or the membrane of the pellicle when being removed from a photomask. For example, this even distribution of the removal force to the pellicle frame may prevent or reduce the likelihood of wrinkling, rupturing, cracking, or other similar defects within a membrane of the pellicle reducing manufacturing costs, reducing waste costs, and preventing or reducing contamination or damage to the photomask to which the pellicle was previously coupled.
(22) FIG. **1**A is a top plan view of a pellicle **100** overlapping a photomask **102** (see FIG. **1**B). The

pellicle **100** includes a pellicle frame **104** that is coupled to the photomask **102**. The pellicle frame **104** includes a first side portion **106**, a second side portion **108** opposite to the first side portion **106**, a third side portion **110** transverse to the first and second side portions **106**, **108** and extends from the first side portion **106** to the second side portion **108**, and a fourth side portion **112** transverse to the first and second side portions **106**, **108**, opposite to the third side portion **110**, and extends from the first side portion **106** to the second side portion **108**. Respective corners of the frame **110** between the first, second, third, and fourth side portions **106**, **108**, **110**, **112** are rounded corners.

(23) The pellicle **100** further includes a membrane **114** that extends across an opening **115** of the pellicle frame **104** defined by the first, second, third, and fourth side portions **106**, **108**, **110**, **112**, respectively. The membrane **114** is configured to, in operation, overlap and cover the photomask **102** such that the membrane **114** acts as a protective cover or boundary preventing or blocking contaminants from reaching the photomask **102**. The membrane **114** may be made of a material that allows selected wavelengths of light to pass through the membrane **114** and reach a target aligned with the membrane **114** and the photomask **102**. For example, the membrane **114** may be made of a material that allows for EUV light to readily pass through the membrane **114** such that the target (e.g., a semiconductor wafer, semiconductor substrate, semiconductor layer, or some other similar semiconductor structure) is utilizing the membrane **114** and the photomask **102**. Alternatively, the membrane **114** may be made of a material that allows for some other wavelength of light to pass through the membrane **114** that may be utilized to carry out various types of patterning techniques.

(24) A dis-pellicle tool or pellicle removal tool includes a first clamp structure **116** and a second clamp structure **118** opposite to the first clamp structure **116**. The first clamp structure **116** includes a first group of pins **120**, and the second clamp structure **118** includes a second group of pins **122**. The first group of pins **120** includes two pins **120**, and the second group of pins **122** includes two pins **122**.

(25) FIG. **1**B is a cross-sectional view of the pellicle **100**, the first clamp structure **116**, and the second clamp structure **118** taken along line **1**B-**1**B′ as shown in FIG. **1**A. As shown in FIG. **1**B, the first group of pins **120** are inserted into first reception openings **124** that extend through the first side portion **106** of the pellicle frame **104**, and the second group of pins **122** are inserted into second reception openings **126** that extend through the second side portion **108**. At least one of the first openings **124** and at least one of the second openings **126** is shown in the cross-sectional view illustrated in FIG. **1**B. The at least one first opening **124** as shown in FIG. **1**B receives a corresponding pin of the first group of pins **120** and the at least one second opening **126** as shown in FIG. **1**B receives a corresponding pin of the second group of pins **122**. In other words, there is a one-to-one relationship between the first group of pins **120** and the first openings **124** such that each respective pin **120** of the first group of pins **120** is inserted into a corresponding first opening of the first openings **124**, and there is a one-to-one relationship between the second group of pins **122** and the second openings **126** such that each respective pin **122** of the second group of pins **122** is inserted into a corresponding second opening of the second openings **126**.

(26) The photomask **102** includes a first side **128** and a second side **130** opposite to the first side **128**. The first side **128** faces away from the pellicle **100**, and the second side **130** faces towards the pellicle **100**.

(27) The frame **104** of the pellicle **100** includes a third side **132** and a fourth side **134** opposite to the third side **132**. The third side **132** faces towards the second side **130** of the photomask **102**, and the fourth side **134** faces away from the photomask **102**. The membrane **114** is coupled to the fourth side **134** of the frame **104** of the pellicle **100**.

(28) An adhesive **136** is on the second side **130** of the photomask **102** and is on the third side **132** of the frame **104**. The adhesive **136** couples or adheres the third side **132** of the frame **104** of the pellicle **100** to the second side **130** of the photomask **102**. The adhesive **136** may a glue or some other type of adhesive material.

(29) In operation, when the first group of pins **120** are within the first openings **124** and the second group of pins **122** are in the second opening **126**, a force **138** is applied to the pellicle frame **104** that is directed away from the photomask **102**. The force **138** may be referred to as a removal force. Based on the orientation as shown in FIG. **1**B, the force **138** is directed in a downward direction. The removal force **138** is strong enough to overcome an adhesion force or strength of which the adhesive **136** adheres or couples together the frame **104** to the photomask. For example, when the removal force **138** is applied to the frame **104**, the adhesive **136** is broken resulting in the removal of the frame **104** from the photomask **102**.
(30) As there are two respective pins **120** in the first group of pins **120** and in the second group of pins **122**, the removal force **138** applied to the frame **104** of the pellicle **100** may be much greater in close proximity to the respective pins of the first group of pins **120** and the second group of pins **122** along the first and second clamp structures **116**, **118**, respectively. In other words, the removal force **138** applied by the first and second clamp structures **116**, **118** at the first and second side portions **106**, **108** may result in stresses and strains in the frame **104** and in the membrane **114** near or adjacent to the first and second side portions **106**, **108** being much greater than those stresses and strains at the third and fourth side portions **110**, **112**. This uneven distribution of the removal force **138** across the frame **104** results in this uneven distribution of stresses and strains in the membrane **114** and may result in flexure, bending, and warpage of the membrane **114**, which may readily be seen in FIG. **1**B. This flexure, bending, and warpage of the membrane **114** may result in propagation of damage or defects within the membrane **114**, which may readily be seen in FIG. **2**.
(31) FIG. **2** is a top plan view a respective pellicle that is the same as the pellicle **100** after being removed from a respective photomask that is the same as the photomask **102**. As shown in FIG. **2**, the membrane **114** of the pellicle has been damaged by removing the pellicle **100** from the photomask **102** utilizing the first and second clamp structures **116**, **118**. As shown in FIG. **2**, a damaged region **140** extending across the membrane **114** is present along locations **142** of the second side portion **108** of the frame **104** of the pellicle **100**. These locations **142** may correspond to, be in close proximity to, or near respective locations at which the removal force **138** was applied to the second side portion **108** to remove the pellicle **100** from the photomask **102**. In other words, these locations **142** may correspond to, be in close proximity to, or near respective locations of the second openings **126** along the second side portion **108** of the frame **104** of the pellicle **100**. While the damaged region **140** propagated at the second side portion **108** of the frame **104** due to bending, flexure, or warpage within the membrane **114**, damage may occur at other regions or locations along the membrane **114** when the removal force **138** is applied to the frame **104** of the pellicle **100** when removing the pellicle **100** from the photomask **102** by overcoming the adhesion force or strength of the adhesive **136**.
(32) FIG. **3**A is a top plan view of a pellicle **200** coupled to the photomask **102**. The pellicle **100** further includes a pellicle frame **202**. The pellicle frame **202** may be made of a metal material. The frame includes a first side portion **204**, a second side portion **206** opposite to the first side portion **204**, a third side portion **208** transverse to the first and second side portions **204**, **206** and extends from the first side portion **204** to the second side portion **206**, and a fourth side portion **210** opposite to the third side portion **208**, transverse to the first and second side portions **204**, **206**, and extends from the first side portion **204** to the second side portion **206**. Respective corners of the frame **202** between the first, second, third, and fourth side portions **204**, **206**, **208**, **210** are rounded corners. The pellicle **200** further includes the membrane **114** that extends across and overlaps an opening **212** of the frame **202** defined by the first, second, third, and fourth side portions **204**, **206**, **208**, **210** of the frame **202**.
(33) A dis-pellicle tool or pellicle removal tool includes a first clamp structure **214**, a second clamp structure **216** opposite to the first clamp structure **214**, a third clamp structure **218** transverse to the first and second clamp structures **214**, **216**, and a fourth clamp structure **220** transverse to the first and second clamp structures **214**, **216** and opposite to the third clamp structure **218**. The first clamp

structure **214** includes a first group of pins **222**, the second clamp structure **216** includes a second group of pins **224**, the third clamp structure **218** includes a third group of pins **226**, and the fourth clamp structure **220** includes a fourth group of pins **228**. In the embodiment of the dis-pellicle tool and the pellicle removal tool as shown in FIG. **3**A, each of the first, second, third, and fourth group of pins **222**, **224**, **226**, **228** includes three pins. In some embodiments, the first, second, third, and fourth group of pins **222**, **224**, **226**, **228** may have different numbers of respective pins relative to each other. For example, the first and second groups of pins **222**, **224** may include three pins whereas the third and fourth groups of pins **226**, **228** may include four pins. In other words, there may be various combinations of numbers of the first group of pins **222**, the second group of pins **224**, the third group of pins **226**, and the fourth group of pins **228**.
(34) FIG. **3**B is a cross-sectional view of the pellicle **200**, the first clamp structure **214**, and the second clamp structure **216** taken along line **3**B-**3**B′ as shown in FIG. **3**A. As shown in FIG. **3**B, the first group of pins **222** are inserted into first reception openings **230** that extend through the first side portion **204** of the frame **202**, and the second group of pins **224** are inserted into second reception openings **232** that extend through the second side portion **206** of the frame **202**. At least one of the first openings **230** and at least one of the second openings **232** is shown in the cross-sectional view illustrated in FIG. **3**B. The at least one first opening **230** as shown in FIG. **3**B receives a corresponding pin of the first group of pins **222**, and the at least one second opening **232** as shown in FIG. **3**B receives a corresponding pin of the second group of pins **224**. In other words, there is a one-to-one relationship between the first group of pins **222** and the first openings **230** such that each respective pin **222** of the first group of pins **222** is inserted into a corresponding first opening **230** of the first openings **230**, and there is a one-to-one relationship between the second group of pins **224** and the second openings **232** such that each respective pin **224** of the second group of pins **224** is inserted into a corresponding second opening **232** of the second openings **232**.
(35) While not shown, the third side portion **208** of the frame **202** includes third reception openings and the fourth side portion **210** includes fourth reception openings. The third and fourth reception openings are the same or similar to the first and second reception openings. The third reception openings of the third side portion **208** receive the third group of pins **226** of the third clamp structure **218**, and the fourth reception openings of the fourth side portion **210** receive the fourth group of pins **228** of the fourth clamp structure **220**. There may be a one-to-one relationship between the third openings and the third group of pins **226** such that each respective pin **226** of the third group of pins **226** is inserted into a corresponding opening of the third openings, and there may be a one-to-one relationship between the fourth openings and the fourth group of pins **228** such that each respective pin **228** of the fourth group of pins **228** is inserted into a corresponding opening of the fourth openings.
(36) The frame **202** of pellicle **200** includes a third side **234** and a fourth side **236** opposite to the third side **234**. The third side **234** faces towards the second side **130** of the photomask **102**, and the fourth side **236** faces away from the photomask **102**.
(37) A first adhesive **238** is on the second side **130** of the photomask **102** and is on the third side **234** of the frame **202** of the pellicle **200**. The first adhesive **238** couples third side **234** of the frame **202** of the pellicle **200** to the second side **130** of the photomask **102**. The first adhesive may be a glue or some other type of adhesive.
(38) A second adhesive **240** is on the fourth side **236** of the frame **202** of the pellicle **200** and couples a membrane support frame **242** to the fourth side **236** of the frame **202** of the pellicle **200**. The second adhesive **240** may be a glue or some other type of adhesive. The membrane **114** is coupled to the membrane support frame **242**.
(39) The membrane support frame **242** may be made of a silicon material. In some embodiments, the membrane support frame **242** may be integral with the membrane **114** such that the membrane support frame **242** and the membrane **114** are made a continuous material, for example, silicon.
(40) In operation, when the first, second, third, and fourth group of pins **222**, **224**, **226**, **228**,

respectively, are inserted into the first reception openings **230** of the first side portion **204**, the second reception openings **232** of the second side portion **206**, the third reception openings (not shown) of the third side portion **208**, and the fourth reception openings (not shown) in the fourth side portion **210**, respectively, the force **138** is applied to the pellicle frame **202** that is directed away from the photomask **102**. Based on the orientation as shown in FIG. **3**B, the force **138** is directed in the downward direction. The removal force **138** is strong enough to overcome an adhesion force or strength of which the first adhesive **238** adheres or couples together the third side **234** of the frame **202** to the second side **130** of the photomask **102**. For example, when the removal force **138** is applied to the frame **202**, the first adhesive **238** is broken resulting in the removal of the frame **202** from the photomask **102**.

(41) As there are more of the respective pins **222**, **224**, **226**, **228** of the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** relative to the respective pins **120**, the force **138** is more evenly distributed across the frame **202** when removing the frame **202** from the photomask **102** relative to when the first and second groups of pins **120**, **122** are utilized to remove the frame **104** from the photomask **102** as discussed earlier herein. The force **138** being more evenly distributed across the frame **202** further prevents or reduces the likelihood of the damage as shown in FIG. **2** from occurring such that manufacturing costs and waste costs are reduced.

(42) As there are more of the respective clamp structures **214**, **216**, **218**, **220** as shown in FIG. **3**B relative to the respective clamp structures **116**, **118** as shown in FIG. **2**, the force **138** is more evenly distributed across the frame **202** when removing the frame **202** from the photomask **102** with the respective clamp structures **214**, **216**, **218**, **220** relative to when the respective clamp structures **116**, **118** are utilized to remove the frame **104** from the photomask **102**. The force **138** being more evenly distributed across the frame **202** further prevents or reduces the likelihood of damage as shown in FIG. **2** from occurring such that manufacturing costs and waste costs are reduced.

(43) As the second adhesive **240** is present and couples the membrane support frame **242** to the fourth side **236** of the frame **202** of the pellicle **200**, the second adhesive prevents or reduces the likelihood of the membrane **114** being exposed to stresses and strains that occur when applying the removal force **138** to the frame **202** utilizing the respective groups of pins **222**, **224**, **226**, **228** and the respective clamp structures **214**, **216**, **218**, **220**. In other words, the second adhesive **240** may be configured to, in operation, act as a stress and strain isolated such that stress and strain that occurs in the frame **202** is isolated from membrane **114** preventing or reducing the likelihood of the damage as shown in FIG. **2** from occurring within the membrane **114**. The second adhesive **240** may have a relatively high elastic modulus for an adhesive material to further isolate the membrane **114** from the any stress or stain that may occur within the frame **202** when applying the force **138** to the frame **202** to remove the frame **202** from the photomask **102**.

(44) As shown in FIG. **3**B, there is less bending, flexure, and warping within the membrane **114** relative to the bending, flexure, and warping of the membrane **114** as shown in FIG. **1**B. The lesser bending, flexure, and warping within the membrane as shown in FIG. **3**B further prevents or reduces the likelihood of damage or defects propagating within the membrane **114**.

(45) FIG. **3**C is a side view of the pellicle **200** coupled to the photomask **102** and the photomask **102** coupled to a photomask stage **244**, which includes one or more photomask alignment structures **246** such that the photomask **102** is properly positioned. The photomask stage **244** includes a stage plate **246** that includes clamp structure openings **248** that are configured to allow the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**, respectively, to pass through a corresponding clamp structure opening **248** such that the first, second, third, and fourth group of pins **222**, **224**, **226**, **228**, respectively, may be inserted into the first reception openings **230** in the first side portion **204**, the second reception openings **232** in the second side portion **206**, the third reception openings (not shown) in the third side portion **208**, and the fourth reception openings (not shown) in the fourth side portion **210**, respectively.

(46) The photomask **102** may be coupled to the photomask alignment structure **246** by an adhesive (not shown). The photomask alignment structure **246** has one or more surfaces **250** that may be adjacent to or may abut the second side **130** of the photomask **102**. When the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228**, respectively, are inserted into the first reception openings **230** in the first clamp structure **214**, the second reception openings **232** in the second clamp structure **216**, the third reception openings (not shown) in the third clamp structure **218**, and the fourth reception openings (not shown) in the fourth clamp structure **220** and the force **138** is applied to the frame **202**, the one or more surfaces **250** may hold the photomask **102** in a stationary position such that the first adhesive **238** breaks resulting in the pellicle **200** being removed from the second side **130** of the photomask **102**.
(47) FIG. **3**D is a zoomed in and enhanced view of section **3**D-**3**D′ as shown in FIG. **3**C of the first side portion **204** of the frame **202** of the pellicle **200** as shown in FIG. **3**C. As shown in FIG. **3**C, adjacent first reception openings **230** are spaced apart by a distance **252**, which extends between center points of the adjacent first reception openings **230**. In at least some embodiments, the distance **252** is determined utilizing the following formulas:
(48) $$\left.\begin{array}{l} f = \frac{M_z}{W_y} = \frac{3FL}{2bd^3} \\ F = F \quad = P \quad \times \end{array}\right\}$$ .fwdarw. $L = 6.498\sqrt{\frac{1}{}}$ σ.sub.f—Silicon bending stress

M.sub.z=Moment relative to Neutral Axis z W.sub.y=Resistance Moment relative to Neutral Axis F=Pellicle bending force with jig pin L=Distance between jig pin b=Sample (pellicle) width h=Sample (pellicle) height P.sub.μ=Sample (pellicle) quality μ=Sample (pellicle glue) adhesion
(49) FIG. **4**A is a top plan view of an embodiment of at least one of the respective pins **222**, **224**, **226**, **228**. As shown in FIG. **4**A, the respective pin **222**, **224**, **226**, **228** includes a first end **254** and a second end **256** opposite to the first end **254**. The first end **254** is an end that is initially inserted into a respective reception opening of the respective reception openings of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**, respectively. The second end **256** is an end that is coupled to the frame **202** of the pellicle **200** such that the respective pin **222**, **224**, **226**, **228** extends and protrudes from at least one of the first, second, third, and fourth side portions **204**, **206**, **208**, **210**, respectively, of the frame **202**. The first end **254** has a first area, and the second end **256** has a second area that is less than the first area of the first end **254**. The respective pin **222**, **224**, **226**, **228** includes at least one inclined surface **258** that extends from the first end **254** to the second end **256**.
(50) FIG. **4**B are various front side views of various embodiments of the respective pin **222**, **224**, **226**, **228**. In the left-most embodiment as shown in FIG. **4**B, the first end **254** has a first circular profile, the second end **256** has a second circular profile larger than the first circular profile, and the at least one inclined surface **258** extends from the first end **254** to the second end **256**. In the center embodiment as shown in FIG. **4**B, the first end has a first oval profile, the second end has a second oval profile larger than the first oval profile, and the at least one inclined surface **258** extends from the first end **254** to the second end **256**. In the right-most embodiment as shown in FIG. **4**B, the first end **254** has a first rectangular profile, the second end **256** has a second rectangular profile larger than the first rectangular profile, and the at least one inclined surface **258** that extends from the first end **254** to the second end **256**. In alternative embodiments, the first and second ends **254**, **256** of the respective pin **222**, **224**, **226**, **228** may have a different polygonal profile than as shown in FIG. **4**B. For example, the first and second ends **254**, **256** may have a triangular profile, an n-gon polygonal profile, or some other similar polygonal or shaped profile.
(51) The first end **254** of the respective pin **222**, **224**, **226**, **228** may have a first width **260** that extends across the first end **254**. In some embodiments, the first width **260** may be greater than or equal to 0.2 millimeters (mm). The second end **256** may have a second width **262** that is greater than the first width **260**. When the respective pin **222**, **224**, **226**, **228** has a circular or oval profile as discussed earlier herein, the first and second widths **260**, **262** may be diameters.
(52) The at least one inclined surface **258** has a slope. In some embodiments, the slope may be

greater than or equal to five (5), may be less than or equal to eight (8), or may be some intermediate value between five (5) and eight (8).

(53) When the respective pin **222**, **224**, **226**, **228** is a respective pin of the first group of pins **222**, the first reception openings **230** may have a width (e.g., a diameter when the first reception openings **230** are circular or oval shaped) and the width may be larger than the first width **260** of the first end **254**. The width of the first reception openings **230** being larger than the first width **260** of the first end **254** allows for the respective pin **222** to be inserted into a corresponding first reception opening **230** of the first reception openings **230**. While this discussion is with respect to a respective pin of the first group of pins **222**, this discussion may readily apply to the second, third, and fourth groups of pins **224**, **226**, **228**, respectively.

(54) FIG. **5**A is a top plan view of a system **300** configured to, in operation, control and position one or more clamps of the dis-pellicle tool or the pellicle removal tool, control and position the stage **244**, or control and position both the one or more clamps of the dis-pellicle tool or the pellicle removal tool and the stage **244**, respectively. The system **300** includes one or more sensors **302***a*, **302***b*, **302***c*. For example, the one or more sensors **302***a*, **302***b*, **302***c* may be a charge-coupled image capture device that captures images by converting photons to electrons.

(55) A controller **304** is in communication with the one or more sensors **302***a*, **302***b*, **302***c*, respectively. The controller **304** may be in communication with the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**, may be in communication with the stage **244**, or may be in communication with both the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** and the stage **244**.

(56) A first sensor **302***a* of the one or more sensors **302***a*, **302***b*, **302***c* monitors the membrane **114** of the pellicle **200** for the propagation of damage or defects within the membrane **114**. For example, if the first sensor **302***a* begins to detect any propagation of damage or defects within the membrane **114** when removing the pellicle **200** from the photomask **102**, the controller **304** may output a control signal to stop movement of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**, to stop movement of the stage **244**, or to stop movement of both the first, second, third and fourth clamp structures **214**, **216**, **218**, **220** and the stage **244** to prevent irreversible damage or defects within the membrane **114** when removing the pellicle **200** from the photomask **102**.

(57) A second sensor **302***b* of the one or more sensors **302***a*, **302***b*, **302***c* may monitor positioning of the first reception openings **230** of the first side portion **204** of the frame **202** relative to the first group of pins **222**. For example, if the second sensor **302***b* of the one or more sensors **302***a*, **302***b*, **302***c* determines that the first reception openings **302***b* are not aligned (e.g., the first group of guide pins **222** are offset relative to the first reception openings **230**) with the first reception openings **230**, the controller **304** may output a control signal to move either one of or both of the first clamp structure **214** and the stage **244** to align the first reception openings **230** with the first group of guide pins **222** such that the first group of guide pins **222** may be inserted into the first reception openings **230**.

(58) A third sensor **302***c* of the one or more sensors **302***a* may monitor positioning of the third reception openings of the third side portion **208** of the frame **202** of the pellicle **200** relative to the third group of guide pins **226**. For example, if the third sensor **302***c* of the one or more sensors **302***a*, **302***b*, **302***c* determines that the third reception openings are not aligned (e.g., the third group of guide pins **226** are offset relative to the third reception openings) with the third reception openings, the controller **304** may output a control signal to move either one of or both of the third clamp structure **218** and the stage **244** to align the third reception openings with the third group of guide pins **226** such that the third group of guide pins **226** may be inserted into the third reception openings of the third side portion **208** of the frame **202** of the pellicle **200**.

(59) While not shown, the controller **304** may be in electrical communication with one or more actuators (not shown) that are in mechanical cooperation or engagement with respective ones of the

first, second, third, and fourth clamp structures **222**, **224**, **226**, **228** and the stage **244**. The controller **304** may send control signals to the one or more actuators (not shown) to control various movements of at least one of the first, second, third, and fourth clamp structures **222**, **224**, **226**, **228**, the stage **244**, or both the first, second, third, and fourth clamp structures **222**, **224**, **226**, **228** and the stage **244**.

(60) While three of the one or more sensors **302**_a_, **302**_b_, **302**_c_ are present in the embodiment of the system **300** as shown in FIG. **5**A, in some alternative embodiments, there may be more than three of the one or more sensors. For example, there may be five sensors, may be six sensors, or may be some greater number of sensors in alternative embodiments of the system **300**. For example, in at least one alternative embodiment, there may be one sensor that monitors the membrane **114** for the onset of the propagation of damage or defects in the membrane **114** and there may be four other sensors for monitoring each of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**.

(61) FIG. **5**B is a zoomed in and enhanced view of section **5**B-**5**B′ of the first clamp structure **214**, one of the respective pins **222** of the first group of pins **222**, and one of the first reception openings **230** of the first reception openings **230** in the first side portion **204** of the frame **202** of the pellicle **200**. As shown in FIG. **5**B, when the respective pin **222** is offset relative to the corresponding first reception opening **230**, either at least one of or both of the first clamp structure **214** and the stage **244** is moved such that a first centerline **306** of the respective pin **222** is aligned with a second centerline **308** of the corresponding first reception opening **230**. Once the first and second centerlines **306**, **308** are aligned with each other, the respective pin **222** as shown in FIG. **5**B may then be inserted into the first reception opening **230** as shown in FIG. **5**B.

(62) FIG. **6** is directed to a cycle flowchart **400** of the pellicle **200** when being utilized within a semiconductor manufacturing factory (FAB). The cycle flowchart **400** includes a first step **402**, a second step **404**, a third step **406**, a fourth step **408**, and a fifth step **410** that are carried out successively and cyclically that the pellicle **200** undergoes multiple times before end of its useful life span.

(63) In a first step **402**, the pellicle **200** is mounted, adhered, bonded, or coupled to the photomask **102** by the first adhesive **238**, which may readily be seen in FIG. **3**B of the present disclosure. After the pellicle **200** is coupled to the photomask by the first adhesive **238**, the photomask **102** may be mounted, adhered, bonded, or coupled to the one or more surfaces **250** of the stage **244**, which may be within a patterning tool (e.g., an EUV patterning tool or some other type of optical patterning tool that may utilize some other wavelength of light).

(64) After the first step **402** in which the pellicle **200** is coupled to the photomask **102** and the photomask **102** is coupled to the stage **244** within the patterning tool, a second step **404** is performed in which a light source emits light at the membrane **114** and the photomask **102** to pattern a target overlapped and aligned with the membrane **114** and the photomask **102**. For example, the target may be a semiconductor substrate, a semiconductor wafer, a semiconductor layer, or some other similar or like type of semiconductor structure or material.

(65) After the second step **404** in which the patterning process has been completed, in a third step **406** the pellicle **200** is removed from the photomask **102** utilizing the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** as shown in FIGS. **3**A and **3**B. The details of removing the pellicle **200** in the third step **406** will be discussed in greater detail with respect to FIGS. **7** and **8**A-**8**E as follows herein.

(66) After the third step **406** in which the pellicle **200** is removed from the photomask **102**, in a fourth step **408** the pellicle **200** is inspected and cleaned such that contaminants that impede light passing through the membrane **114** are removed from the membrane **114**.

(67) After the fourth step **408** in which the membrane **114** of the pellicle **200** is cleaned, in a fifth step **410** the membrane **114** of the pellicle **200** and the photomask are inspected for damage and defects to determine whether the pellicle **200** and the photomask may be reutilized in the patterning

process to pattern a new target (e.g., a new semiconductor substrate, a new semiconductor wafer, a new semiconductor layer, or a new semiconductor structure or material) within the patterning tool.
(68) After the fifth step **410** in which the membrane **114** of the pellicle **200** and the photomask **102** are inspected, if both the pellicle **200** and the photomask **102** pass inspection, the respective steps **402**, **404**, **406**, **408**, **410** of the cycle flowchart **400** are performed again to reutilize the pellicle **200** and the photomask **102**.
(69) FIG. **7** is a flowchart of additional details with respect to the third step **406** in the cycle flowchart **400**. The flowchart includes respective steps **406***a*, **406***b*, **406***c*, **406***d*.
(70) In a first step **406***a*, the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** of the dis-pellicle tool or pellicle removal tool are moved into position around the stage **244**. The first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** may be positioned around the stage **244** such that each one of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are aligned with a corresponding clamp structure opening **248** within the stage **244** (see FIG. **3**D of the present disclosure). For example, in the first step **406***a*, the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** may be moved from a lowered position to a raised position in a first direction **500** (see FIG. **8**D) by the one or more actuators as discussed earlier herein. Once the first step **406***a* is performed, the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are positioned as shown in FIG. **8**A and FIG. **8**D such that the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are around the pellicle **200** and are level with the pellicle **200**.
(71) After the first step **406***a* in which the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are positioned around the pellicle **200** and are raised to be level with the pellicle **200**, at least one of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are moved, the stage **244**, or both the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** and the stage **244** are moved such that the respective reception openings in the first, second, third, and fourth side portions **204**, **206**, **208**, **210** are aligned with corresponding ones of the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220**. As shown in FIG. **8**A, the stage **244** is moved in a second direction **502** such that the stage **244** moves the photomask **102** and the pellicle diagonally based on the orientation as shown in FIG. **8**A. While the stage **244** is shown as moving in FIG. **8**A, in some alternative embodiments, the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** may be individually moved to align the respective reception openings in the first, second, third, and fourth side portions **204**, **206**, **208**, **210** with the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228**. While the stage **244** is shown as moving in FIG. **8**A, in some alternative embodiments, both the stage **244** may be moved and the first, second, third, and fourth side portions **204**, **206**, **208**, **210** may be individually moved and adjusted or moved and adjusted together to align the respective reception openings in the first, second, third, and fourth side portions **204**, **206**, **208**, **210** with the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** of the first, second, third and fourth clamp structures **214**, **216**, **218**, **220**.
(72) After the second step **406***b* in which the respective reception openings in the first, second, third, and fourth side portions **204**, **206**, **208**, **210** are aligned with the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228**, in a third step **406***c* the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** are moved inward toward the frame **202** of the pellicle **200** to insert the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** of the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** into the respective reception openings in the first, second, third, and fourth side portions **204**, **206**, **208**, **210** of the frame **202**. This inward movement to insert the respective pins into the reception openings is represented by arrows **504** as shown in FIG. **8**B. The first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** being inserted into the respective reception openings of the first, second, third, and fourth side portions **204**, **206**, **208**, **210**, respectively, may readily be seen in FIG. **8**C and FIG. **8**E.
(73) After the third step **406***c* in which the first, second, third, and fourth groups of pins **222**, **224**,

226, **228** have been inserted into the respective reception openings of the first, second, third, and fourth side portions **204**, **206**, **208**, **210** of the frame **202** of the pellicle **200**, in a fourth step **406***d* the removal force **138** is applied to the frame **202** by moving the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** away from the photomask **102**, which is directed in the downward direction based on the orientation as shown in FIG. **8**E. As the first, second, third, and fourth clamp structures **214**, **216**, **218**, **220** move away from the photomask the first, second, third, and fourth groups of pins **222**, **224**, **226**, **228** pull downward on the frame **202** such that the first adhesive **238** is broken removing the pellicle **200** from the photomask **102**. After the pellicle **200** is removed from the photomask **102**, the other steps of the cycle flowchart **400** may be performed. The process as described with respect to the flowchart **500** may be performed over and over again as the cycle flowchart **400** is performed over and over again.

(74) While not discussed in detail herein, the respective reception openings of the first, second, third, and fourth side portions **204**, **206**, **208**, **210** of the frame **202** of the pellicle **200** may be configured to, in operation, allow for venting of fluid between the membrane **114** and the photomask **102** when performing a patterning process on a semiconductor wafer, a semiconductor substrate, a semiconductor layer, or some other similar or like type of semiconductor material.

(75) As is readily apparent in view of the above discussion herein, the present disclosure provides one or more embodiments of a dis-pellicle tool or pellicle removal tool including one or more clamp structures that are utilized to apply a removal force to a pellicle frame of a pellicle in order to remove the pellicle from a photomask. When the force is applied by the one or more clamp structures to the pellicle frame, the force is applied in a balanced manner across the pellicle frame such that the force is evenly distributed across and along the pellicle frame. This even distribution of the removal force prevents or reduces the likelihood of damage to the pellicle or the membrane of the pellicle when being removed from a photomask. For example, this even distribution of the removal force to the pellicle frame may prevent or reduce the likelihood of wrinkling, rupturing, cracking, or other similar defects within a membrane of the pellicle reducing manufacturing costs, reducing waste costs, and preventing or reducing contamination or damage to the photomask to which the pellicle was previously coupled.

(76) Preventing or reducing the likelihood of damage or defects propagating within a membrane of a pellicle reduces manufacturing costs, wastes costs, and increases the usable life span of the pellicle. The increased life span of the pellicle reduces replacement costs as the pellicle may be utilized several more times before needing to be replaced due to damage or defects propagating within the membrane of the pellicle.

(77) At least one embodiment of a pellicle frame of the present disclosure may be summarized as including: a frame including: a first surface; a second surface opposite to the first surface; a first side portion; a second side portion opposite to the first side portion; a third side portion transverse to the first and second side portions, the third side portion extends from the first side portion to the second side portion and is coupled to the first side portion and the second side portion; a fourth side portion transverse to the first and second side portions, the fourth side portion extends from the first side portion to the second side portion and is coupled to the first side portion and the second side portion, and the fourth side portion is opposite to the third side portion; an opening delimited by the first, second, third, and fourth side portions; a first group of holes in the first side portion; a second group of holes in the second side portion; a third group of holes in the third side portion; and a fourth group of holes in the fourth side portion.

(78) At least one embodiment a method of the present disclosure may be summarized as including: monitoring a first clamp structure with a first group of pins, a second clamp structure with a second group of pins, a third clamp structure with a third group of pins, and a fourth clamp structure with a fourth group of pins with a plurality of image sensors; aligning the first group of pins of the first clamp structure with a first group of holes in a first side portion of a pellicle frame of a pellicle; aligning the second group of pins of the second clamp structure with a second group of holes in a

second side portion of the pellicle frame opposite to the first side portion; aligning the third group of pins of the third clamp structure with a third group of holes in a third side portion of the pellicle frame transverse to the first and second side portions; aligning the fourth group of pins of the fourth clamp structure with a fourth group of holes in a fourth side portion of the pellicle frame opposite to the third side portion and transverse to the first and second side portions; inserting the first, second, third, and fourth group of pins in the first, second, third, and fourth groups of holes, respectively; and applying a force to the pellicle frame through the first, second, third, and fourth groups of pins in the first, second, third, and fourth groups of holes, respectively, by moving the first, second, third, and fourth clamp structures away from a photomask coupled to the pellicle frame while maintaining the photomask in a stationary position.

(79) At least one embodiment of a dis-pellicle tool may be summarized as including: a pellicle clamp structure configured to, in operation, clamp onto a pellicle frame of a respective pellicle coupled to a photomask, the pellicle clamp structure including: a body portion; one or more tapered pins that protrudes from the body portion, the one or more tapered pins includes: a first end; a second end opposite to the first end; and at least one inclined sidewall that is transverse to the first end and the second end, the at least one inclined sidewall extends from the first end to the second end.

(80) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A pellicle frame, comprising: a frame including: a first surface; a second surface opposite to the first surface; a first side portion; a second side portion opposite to the first side portion; a third side portion transverse to the first and second side portions, the third side portion extends from the first side portion to the second side portion and is coupled to the first side portion and the second side portion; a fourth side portion transverse to the first and second side portions, the fourth side portion extends from the first side portion to the second side portion and is coupled to the first side portion and the second side portion, and the fourth side portion is opposite to the third side portion; an opening delimited by the first, second, third, and fourth side portions; a group of first holes in the first side portion, the group of first holes includes a first number, and the group of first holes are empty; a group of second holes in the second side portion, the group of second holes includes a second number, and the group of second holes are empty; a group of third holes in the third side portion, the group of third holes includes a third number, and the group of third holes are empty; and a group of fourth holes in the fourth side portion, the group of fourth holes includes a fourth number, and the group of fourth holes are empty, and wherein the first number, the second number, the third number, and the fourth number are equal to each other, and wherein the group of first holes, the group of second holes, the group of third holes, and the group of fourth holes are configured to, in operation, receive corresponding tapered pins of a plurality of clamping structures.
2. The pellicle frame of claim 1, wherein the groups of the first, second, third, and fourth holes are configured to, in operation, receive corresponding groups of pins of one or more corresponding clamps.
3. The pellicle frame of claim 1, wherein the first number, the second number, the third number,

and the fourth number are greater than or equal to 2.
4. The pellicle frame of claim 1, wherein the first number, the second number, the third number, and the fourth number are equal to each other.
5. The pellicle frame of claim 1, wherein respective adjacent holes of the respective groups of first holes, second holes, third holes, and fourth holes are spaced equidistant from each other.
6. The pellicle frame of claim 5, wherein the respective adjacent holes of the respective groups of first, second, third, and fourth holes are spaced apart from each other to avoid rupturing of a membrane coupled to the frame when the pellicle frame is being removed from a respective photomask by a dis-pellicle tool.
7. The pellicle frame of claim 6, wherein the respective adjacent holes of the first, second, third, and fourth groups of holes are spaced apart from each other by a length determined by a formula as follows:
$L=6.498*\mathrm{sqrt}(1/\mu)$ wherein L is length and μ is an adhesion force of an adhesive coupling the frame of the pellicle to the respective photomask.
8. A method, comprising: monitoring a first clamp structure with a first group of pins, a second clamp structure with a second group of pins, a third clamp structure with a third group of pins, and a fourth clamp structure with a fourth group of pins with a plurality of image sensors; aligning the first group of pins of the first clamp structure with a first group of holes in a first side portion of a pellicle frame of a pellicle; aligning the second group of pins of the second clamp structure with a second group of holes in a second side portion of the pellicle frame opposite to the first side portion; aligning the third group of pins of the third clamp structure with a third group of holes in a third side portion of the pellicle frame transverse to the first and second side portions; aligning the fourth group of pins of the fourth clamp structure with a fourth group of holes in a fourth side portion of the pellicle frame opposite to the third side portion and transverse to the first and second side portions; inserting the first, second, third, and fourth group of pins in the first, second, third, and fourth groups of holes, respectively; and applying a force to the pellicle frame through the first, second, third, and fourth groups of pins in the first, second, third, and fourth groups of holes, respectively, by moving the first, second, third, and fourth clamp structures away from a photomask coupled to the pellicle frame while maintaining the photomask in a stationary position.
9. The method of claim 8, wherein applying the force to the pellicle frame further includes overcoming an adhesion force of an adhesive coupling the pellicle frame to the photomask.
10. The method of claim 9, further comprising, after removing the pellicle frame from the photomask, cleaning a membrane of the pellicle coupled to the pellicle frame of the pellicle.
11. The method of claim 10, further comprising, after cleaning the membrane of the pellicle, coupling the pellicle frame to at least one of the following of the photomask or another photomask.
12. The method of claim 11, wherein inserting the first, second, third, and fourth groups of pins into the first, second, third, and fourth groups of holes, respectively, occurs simultaneously after each of the first, second, third, and fourth groups of pins are aligned with the first, second, third, and fourth groups of holes, respectively.
13. The method of claim 11, wherein aligning the first, second, third, and fourth groups of pins with the first, second, third, and fourth groups of holes further includes: moving a stage structure that supports the photomask.
14. The method of claim 11, wherein aligning the first, second, third, and fourth groups of pins with the first, second, third, and fourth groups of holes further includes: moving at least one of the first, second, third, and fourth clamp structures.
15. A dis-pellicle tool, comprising: a first pellicle clamp structure configured to, in operation, engage with a first portion of a pellicle frame of a respective pellicle coupled to a photomask, the first pellicle clamp structure including: a first body portion; one or more first tapered pins that protrude from the first body portion, the one or more first tapered pins are configured to, in operation, be inserted into one or more first holes along a first side portion of the pellicle frame,

each respective first tapered pin of the one or more first tapered pins includes: a first end; a second end opposite to the first end; and at least one first inclined sidewall that is transverse to the first end and the second end, the at least one first inclined sidewall extends from the first end to the second end; a second pellicle clamp structure configured to, in operation, engage with a second portion of the pellicle frame of the respective pellicle coupled to the photomask, the second portion of the pellicle frame being opposite to the first portion of the pellicle frame, the second pellicle clamp structure including: a second body portion; one or more second tapered pins that protrude from the second body portion, the one or more second tapered pins are configured to, in operation, be inserted into one or more second holes along a second side portion of the pellicle frame opposite to the first side portion of the pellicle frame, each respective second tapered pine of the one or more second tapered pins including: a third end; a fourth end opposite to the third end; and at least one second inclined sidewall transfer to the third end and the fourth end, the at least one second inclined sidewall extends from the third end to the fourth end.
16. The dis-pellicle tool of claim 15, further comprising: a third pellicle clamp structure transverse to the first and second pellicle clamp structures, the third pellicle clamp structure is configured to, in operation, engage with a third portion of the pellicle frame of the respective pellicle coupled to the photomask, the third portion of the pellicle frame being transverse to the first portion of the pellicle frame and transverse to the second portion of the pellicle frame, and the third portion of the pellicle frame extends from the first portion of the pellicle frame to the second portion of the pellicle frame; and a fourth pellicle clamp structure opposite to the third pellicle clamp structure and transverse to the first and second pellicle clamp structures, the fourth pellicle clamp structure is configured to, in operation, engage with a fourth portion of the pellicle frame of the respective pellicle coupled to the photomask, the fourth portion of the pellicle frame being transverse to the first portion of the pellicle frame and transverse to the second portion of the pellicle frame, the fourth portion of the pellicle frame being opposite to the third portion of the pellicle frame, and the fourth portion of the pellicle frame extends from the first portion of the pellicle frame to the second portion of the pellicle frame.
17. The dis-pellicle tool of claim 16, wherein the first, second, third, and fourth pellicle clamp structures are movable in a direction laterally inwards and outwards relative to a central region defined by the first, second, third, and fourth pellicle clamp structures.
18. The dis-pellicle tool of claim 16, wherein the first pellicle clamp structure, the second pellicle clamp structure, the third pellicle clamp structure, and the fourth pellicle clamp structure are configured to, in operation, be individually movable relative to each other and relative to the pellicle frame.
19. The dis-pellicle tool of claim 15, wherein: the first end has a first end surface with a first surface area; and the second end has a second end surface with a second surface area greater than the first surface area.
20. The dis-pellicle tool of claim 15, wherein the first pellicle clamp structure, and the second pellicle clamp structure are configured to, in operation, be individually movable relative to each other and relative to the pellicle frame.